



UPPER LOBE	TB, CF (Cystic Fibrosis), ABPA, Post Radiation
MIDDLE LOBE	NTM (non-tuberculosis mycobacterium), Kartagener Syndrome
LOWER LOBE	Scleroderma, Hypogammaglobulinemia

CASE SCENARIO

A 35-year-old former smoker presents with SOB, productive cough, purulent bronchorrhea whose amount **varies with position of body**. He has lost weight in last year. Examination shows digital clubbing and bilateral scattered crackles heard over lower lobes of lungs.

- 1. Chronic bronchitis:** Smoker, Purulent Bronchorrhea, Rhonchi (I + E), FEV1 ↓ → salbutamol administration → ↓ FEV1 no change (<12%)
- 2. Bronchiectasis**

WORK UP

- CXR → Normal, **tram track appearance**
- HRCT →
 - Signet ring appearance
 - Tram Track appearance
- Pulmonary Function Test → ↓ FEV1, FEV1/FVC <0.7
- Sputum Culture and Microscopy → **Dittrich plugs**
- HIV
- Autoimmune panel
- Alpha 1 antitrypsin panel
- PCR for CFTR gene/ sweat chloride → CF: Mucoviscidosis
- Nasal biopsy/ video microscopic analysis of cilia motility → Kartagener Syndrome
- Quantitative IgG levels

TREATMENT

1. Non tuberculosis mycobacteria T/t: Macrolides + Rmp (Rifampin) + Ethambutol
2. Intermittent antibiotics use: FQ for 2 weeks and 2 weeks off (Ciplox)
3. Mucolytics for Purulent Bronchorrhea (Dornase, DNAase, N-Acetylcysteine, 3% saline Nebulization)
4. For inspissated secretions: Percussion device (High-frequency oscillatory device)



High Frequency Chest Oscillatory Device



Incentive spirometer



Q. Which of the following causes of Bronchiectasis involves the upper lobes?

- a. Kartagener Syndrome
- b. Interstitial lung diseases
- c. Scleroderma
- d. Tuberculosis

Ans (d)



PNEUMONIA

00:00:08

PATHOLOGY

00:13:45

- Organism inhaled → extensive damage → bacteria enters lungs → neutrophilic changes and RBCs are seen.
- Alveolar Edema: Exudate in alveoli
- Red Hepatization: Red blood cells plus PMN infiltrate is present. Spongy texture of lung changes to that of liver that is why it is called red hepatization
- Grey Hepatization: RBC lysis with desquamated cells, fibrin and PMN cell infiltrate.
- Resolution: Alveolar macrophage enter and phagocytose bacteria laden neutrophils

LEADING ORGANISM LEADING TO PNEUMONIA

00:16:05

Community acquired pneumonia (Outpatients)	Pneumococcus > Mycoplasma pneumoniae
Community acquired pneumonia (ICU) CURB 65 score = 3	Pneumococcus
Community acquired pneumonia (Non-ICU) CURB 65 score = 2	Pneumococcus
AIDS patients	Pneumococcus
Alcoholic Alcoholic with red currant jelly sputum	• Klebsiella pneumoniae/ Friedlander's Pneumonia • CXR shows Bulging fissure sign
COPD	Haemophilus influenzae
Hospital acquired pneumonia Nosocomial: > 48 hours of admission	Staphylococcus aureus
Ventilator-associated pneumonia	MDR: Pseudomonas (aeruginosa), Gram negative Bacilli, Acinetobacter Non MDR: Pneumococcus
Dementia, stroke	Anaerobes
Superadded bacterial Pneumonia in influenza A	Staphylococcus aureus
Viral pneumonia with CXR showing Peripheral GGO in lower lobes	SARS-COV-2 (RT PCR for confirmation)
Atypical pneumonia with SIADH	Legionella pneumophila: legionnaire's pneumonia



	<10 years: CA-MRSA >10 years: Pseudomonas aeruginosa, Burkholderia cepacia
Exposure to bats	Histoplasmosis
Exposure of birds	Histoplasma and Chlamydia psittaci
Exposure to Rabbits	Tularemia (Hunter by profession – Eschar formation, Ulceroglandular fever)
Exposure to sheep and goats	Coxiella Burnetti (Q fever)
Stay in hotel/ luxury cruise	Legionella pneumophila (Inhalation of aerosol bacteria, water-borne/ shower in a ship)
Primary lung abscess (with risk factors for aspiration)	Anaerobes
Secondary lung abscess (associated with low immunity)	CA-MRSA (community-acquired MRSA)
Lung abscess	CA-MRSA
Walking pneumonia in winter months	Mycoplasma Pneumoniae
Atypical pneumonia features • Clinico-radiological mismatch • Diarrhea • Confusion • Seizures (due to hyponatremia)	• Mycoplasma pneumoniae • Legionella pneumophila • Chlamydia • SARS-COV-2

CLINICAL FEATURES

00:34:04

SYMPTOMS

- Fever
- Cough and excessive sputum production (Rusty sputum, red currant jelly sputum)
- Chest pain - Pleuritis
- Shortness of breath
- Headache - Hypoxia
- Hemoptysis present – CA-MRSA
- Diarrhea, confusion, hyponatremic seizures - Legionella Pneumophila

Important Information

- Staphylococcus aureus radiological finding is **pneumatocele** (small cavity rupture and communicate with pleural space and can contribute to secondary pneumothorax)



Red currant jelly sputum
(Klebsiella pneumonia)



Rusty sputum (Pneumococcal pneumonia)

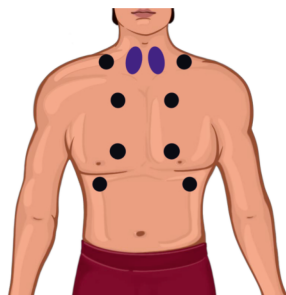
ON EXAMINATION

00:36:59

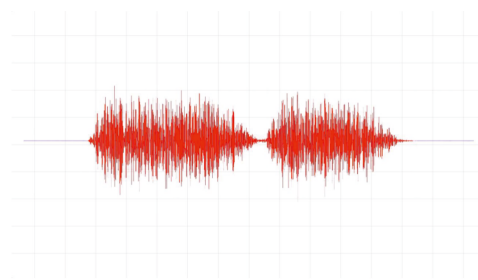
- Respiratory rate: ↑, accessory muscles of respiration (Sternocleidomastoid, Scalaneous, Alae nasi) working overtime
- Subcostal Recession: Present (lower part of chest wall is sucked inside because diaphragm is not contracting with more force to aid with breathing)
- Tactile Fremitus: ↑ (lung tends to become more solidified & the sound conducts better in solid)
- Percussion note: Dull
- Air Entry: Normal
- **Bronchial Breathing** : Presence of pus in alveoli change character of breathing (Presence of very clear audible gap between inspiration & expiration) , (In normal vesicular breathing → no audible gap between inspiration and expiration → soft, rustling of leaf's like breathing)
 - Most reliable auscultation – Bronchial breathing
- Crackles: Present (not reliable)
- Bronchophony/ whispering pectoriloquy
- Egophony – E will be heard as AAA (Nasal component will be present to the sound auscultating on the lung fields of the patient)

BRONCHIAL SOUND

00:42:01



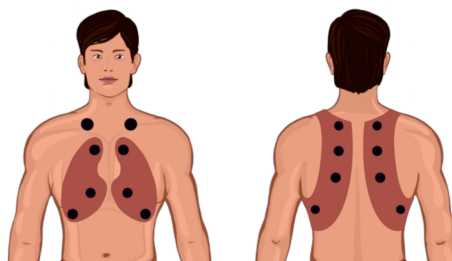
Location where heard normally
manubrium sterni and posteriorly
between C7 and T3 vertebrae



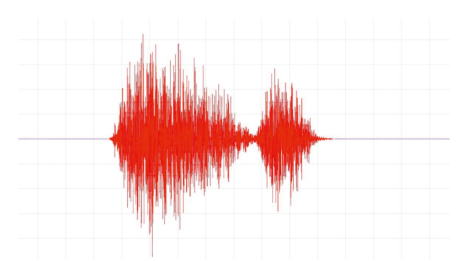
Audible gap

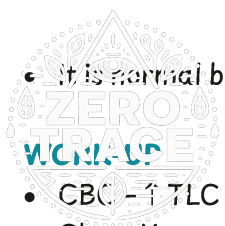
- Pneumococcal pneumonia produces a toxin called **pneumolysin** → activation of platelets → triggering myocardial infarction

VESICULAR SOUNDS



Locations where heard normally in B/L lung fields





- It is normal breathing. There is no gap seen between graph of inspiration and expiration

00:45:19

- CBC - ↑ TLC count
- Chest X-ray: A patch → consolidation in a round format AKA Lobar pneumonia → shows parapneumonic effusion, if not treated → empyema (pus developed in pleural cavity)
- Blood Culture: To demonstrate which organism is causing
- Sputum evaluation : (Criteria for adequacy of sample: > 25 PMN and < 10 squamous epithelial cells)
 - Culture
 - Gram Stain
- Urine Antigen: Legionella & Pneumococcus
- PCR: Cost & availability to be evaluated
- Biomarkers: CRP, Procalcitonin (Indicator for Sepsis), copeptin, STREM-1

CURB-65 SCORING OR UPDATED CRB-65 SCORE

(TO DECIDE FOR NEED FOR HOSPITALIZATION)

PARAMETERS

- Confusion (abbreviated mental test score <8)
- BUN > 7 mmol/L
- R/R >30/min
- BP <90/60 mm Hg
- Age > 65 years

NOTE:

Interpretation of each of these parameters is that they have one point each. Roadblock is BUN report which comes late & without that we can't proceed. So as per updated CRB-65 score all components are same except for BUN value as per final scores are concerned interpretations does not change

SCORE	MORTALITY	DECISION FOR OUTPATIENT AND INPATIENT TREATMENT
0	1.5%	Outpatient
1-2 Age >65 years alone may not need hospitalization	9.2%	Inpatient
≥3	22%	ICU based care

- Clinical features to look out for
 - Inability to take fluids, mental status to enable compliance with medication, spo2 <92% room air

CRITERIA FOR SEVERITY OF CAP

MAJOR CRITERIA	MINOR CRITERIA
• Respiratory failure requiring Mechanical ventilation	• RR >30/min



- Septic shock requiring vasopressors

- $\text{PaO}_2 / \text{FiO}_2 < 250$
- Multilobar infiltrates
- Confusion - Hypoxia
- BUN - $\uparrow$
- TLC - $\downarrow$
- Platelets - $\downarrow$
- Core Temp - $\downarrow$
- Hypotension needing aggressive IVF

TREATMENT

OUTPATIENT BASIS: (If CURB 65 score = 0)

No Comorbidities or risk factors for antibiotic resistance (antibiotics t/t in last 3 months)	With Comorbidities or risk factors for antibiotic resistance (antibiotic t/t in last 3 months)
• Amoxicillin + Macrolides / Doxycycline	• Amoxicillin + Clavulonic acid + Macrolides / Doxycycline or Moxifloxacin

IN-PATIENT BASIS TREATMENT (IF CURB 65 SCORE IS 2)

SEVERE	NON-SEVERE
• No risk factors - Beta Lactam + Macrolides / Beta Lactam + Respiratory Fluoroquinolones • Prior respiratory isolation/ prior antibiotic t/t: Add coverage for MRSA / Pseudomonas aeruginosa ○ For MRSA - Linezolid ○ For Pseudomonas Aeruginosa - Piperacillin - Tazobactam	• IV Respiratory Fluoroquinolones + Beta Lactam (Ampicillin- Sulbactam) / Macrolides

EMPIRICAL ANTIBIOTICS FOR HOSPITAL-ACQUIRED AND VENTILATOR-ASSOCIATED PNEUMONIA

NO RISK FACTORS FOR RESISTANCE IN GNB	RISK FACTORS FOR RESISTANCE IN GNB
• Piperacillin Tazobactam	• Piperacillin Tazobactam + • Amikacin / Levofloxacin / Colistin / Polymyxin-B

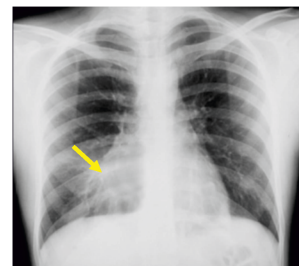
- Add linezolid if MRSA risk factors exist

CASE SCENARIO/ PRESENTATION

Q. 30-year-old patient presents with complaints of cough, productive sputum and shortness of breath. On examination RR 26/min with SpO_2 90% room air. On percussion dullness is noted in right inflammatory area and infra-axillary area. On Auscultation on these sites bronchial breathing is heard. Past medical history reveals patient is HIV positive and has CD4 count of 55 cells/cu.mm. Which is the incriminating organism?

- Pneumocystis jiroveci
- Pneumococcus
- Klebsiella pneumoniae
- Mycoplasma Pneumoniae

Ans: (b)



Lobar Pneumonia

Q.50-year-old man is admitted with complaints of cough and shortness of breath for past 3 days. On examination he appears confused and disoriented. Work up shows spO_2 93% at room air, RR 20/min with pulse 100/min and BP 130/80 mmHg. Work up shows increased TLC, sputum gram stain does not show any organisms, CRP is elevated, BUN is 15 mg/dl, creatinine 0.8 mg/dl, electrolytes: Na 120 meq/L and K 5.2 meq/L. CXR is shown. Which organisms is likely to be responsible?

- Pneumococcus
- COVID-19
- Legionella pneumophila
- Pseudomonas

Ans: (c)



Round Pneumonia

Q.10-year-old child is admitted with diagnosis of pneumonia. CXR is shown below. Which of the following organism is likely to be responsible?

- SARS COV 2
- Anaerobes
- Pneumococcus
- H3N1

Ans: (c)

Q.20-year-old college student on festival consumed lots of alcohol and drugs. Due to intoxication, he kept on sleeping for 2 days. When he woke up, he had halitosis, high grade fever with breathing difficulty and purulent bronchorrhea. CXR was done and showed air fluid level in right middle lobe. Which of the following organism is incriminated and what is best treatment?

- Anaerobes, Clindamycin



- b. *Streptococcus pneumoniae*, Levofloxacin
- a. *Mycoplasma pneumoniae*, Azithromycin
- c. *Pseudomonas*, Piperacillin – Tazobactam

Ans: (a)



Q. 75-year-old man T2DM and CKD is admitted to ICU with diagnosis of Severe pneumonia. CRB score is 3. Which is best treatment of this patient?

- a. IV Moxifloxacin
- b. IV Ceftriaxone plus Azithromycin
- c. IV Moxifloxacin plus ampicillin sulbactam
- d. IV Vancomycin with Piperacillin with Tazobactam

Ans: (c)

Q. Which of the following organisms is not a cause of atypical pneumonia?

- a. *Mycoplasma pneumoniae*
- b. *Legionella pneumophila*
- c. *Chlamydia pneumoniae*
- d. *Pneumococcal pneumonia*

Ans: (d)



BRONCHIAL ASTHMA



SOFT MIST INHALER (SMI)

- Battery powered device
- Medicine - Formoterol, corticosteroid
- **Easiest to use** (especially for geriatric and paediatric patient)
- Easy to carry
- Final lung delivery of medication - 40-50%



METERED DOSE INHALER

- Also called pMDI (pressurised MDI)
- Disadvantage -
 - 1st reflex after pressing canister → exhalation
 - Medicine gets deposited in upper part of airways/oral cavity
- Requires hand-breath co-ordination
- Final lung delivery of medication - 10-20%



DPI

- Dry powder inhaler
- Advantage over pressurized metered dose inhaler - Breath actuated device
- Final lung delivery of medication - 20-40%



- Ideal/optimum particle size delivery to reach terminal bronchiole - 1-3 μm

STEPS FOR CORRECT PMDI (PRESSURISED METERED DOSE INHALER- DOSE INHALER) USE

- Shake the inhaler well (for most formulations): **5-10** seconds to mix medication
- Exhale fully - Breathe out completely to empty your lungs before inhaling medication
- Hold the inhaler upright - Place mouthpiece in mouth with lips sealed around it (or use spacer if recommended)
- Press the inhaler and inhale slowly - As you press canister, take a slow deep breath for 3-5 seconds
- Hold your breath for 10 seconds - This allows medication to settle in lungs
- Exhale slowly - Breathe gently through nose/mouth
- Wait for 30-60 seconds if second dose is needed, then repeat steps
- Rinse your mouth (if using an ICS inhaler)- Prevents oral thrush

VALVED HOLDING CHAMBER

- Patient can inhale and exhale at his own will
- Good compliance of patient
- Better drug delivery to lungs of patient



AIR (ANTI-INFLAMMATORY RELIEVER)	MART (MAINTENANCE AND RELIEVER THERAPY)
• Used in GINA guidelines step 1-2 • Low dose ICS +Formoterol	• Used in GINA guidelines steps 3-4-5 • To be taken daily

BRONCHIAL ASTHMA (BA)

PEAK EXPIRATORY FLOW RATE METER

- Self-assessment device for Asthma
- Normal FEV1
 - FEV1 (Forced expiratory volume in 1 second) >80%
 - FEV2 (Forced Expiratory volume in 2 seconds) >95%

Asthmatic → ↓ FEV1 → Administer Salbutamol and check after 15 mins → FEV1 normalizes

- Increment in FEV1 post bronchodilator - >12% or 200 ml of air
- FEV1 increment post bronchodilator <12% - COPD
- Major diagnostic criteria for BA - Reversibility of airflow obstruction



Peak expiratory flow rate meter

00:13:04

GRADING OF ASTHMA

SEVERITY	FEV1 VALUE
Mild	70-80%
Moderate	40-69%
Severe	<40%
Imminent respiratory arrest	<25%

- Mild, moderate & severe BA - ↓ PO₂, ↓PCO₂ due to hyperventilation

ASTHMA PHENOTYPES

TYPE 2 INFLAMMATION	NON TYPE 2 INFLAMMATION
• Eosinophilic inflammation driven by IL-4, IL-5, IL-13 • Th2 cells, mast cells & Eosinophils • Lab test supporting Type 2 inflammation ◦ Fe NO (Fraction of exhaled Nitric oxide)>25-50 ppb in adults and >20 ppb in children ◦ Blood eosinophil count >150 cells/μl ◦ Sputum eosinophil count >2-3%	• Driven by IL-6, TGF-β • Th1, Th17, Neutrophils

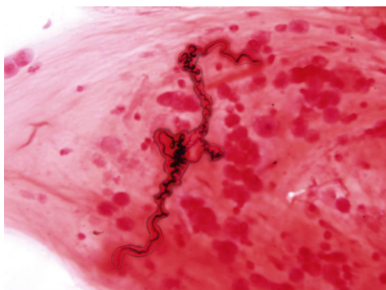


Important information

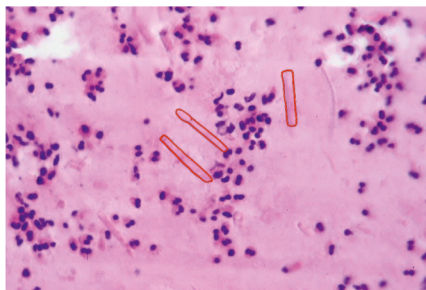
- Nitric Oxide/Vasoreactivity test - used for Pulmonary artery HTN

PATHOLOGICAL FINDINGS IN ASTHMA

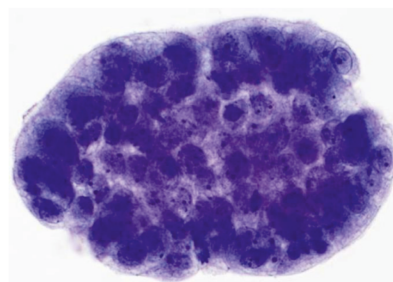
- Curschmann's spirals- Mucus plugs in airways
- Charcot Leyden crystals- Eosinophil proteins (bipyramidal shaped)
- Creola bodies - Sloughed epithelium
- **Thickening** of basement membrane of alveoli
- Hypervascularization of airway
 - Compensatory mechanism ensuring more surface area
 - Diffusion capacity of lung for Carbon monoxide (DLCO) - elevated



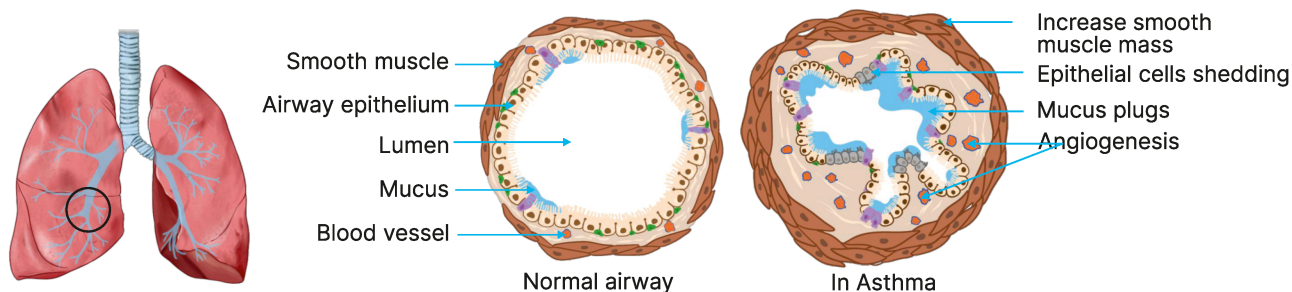
Curschmann Spiral



Charcot Leyden Crystals



Creola bodies



FINDINGS IN ASTHMATICS

- Severe bronchospasm
- Expiratory Ronchi in early stage
- Inspiratory + Expiratory Ronchi in late stage
- Mucus production obstructing airways
- Silent chest as disease progresses to Status asthmaticus

TRIGGERS

- Allergens/irritants : Pollen, dust, House-mites, Cat fur
- Viral infections : RSV, COVID-19, H₃N₁
- Exercise, cold air : Airway hyper-reactivity
- Air pollution : CO, NO, SO₂ (diesel)
- Drugs: Aspirin (COX-1 inhibitors - Inhibit PG production & ↑ levels of leukotrienes), Timolol eye drops, Propranolol
- Occupational exposure : Cleaning staff, Nursing staff

00:32:03



COMORBIDITIES THAT MUST BE ASSESSED IF ATTACKS OCCUR IN SPITE OF MODERATE DOSE ICS & LABA

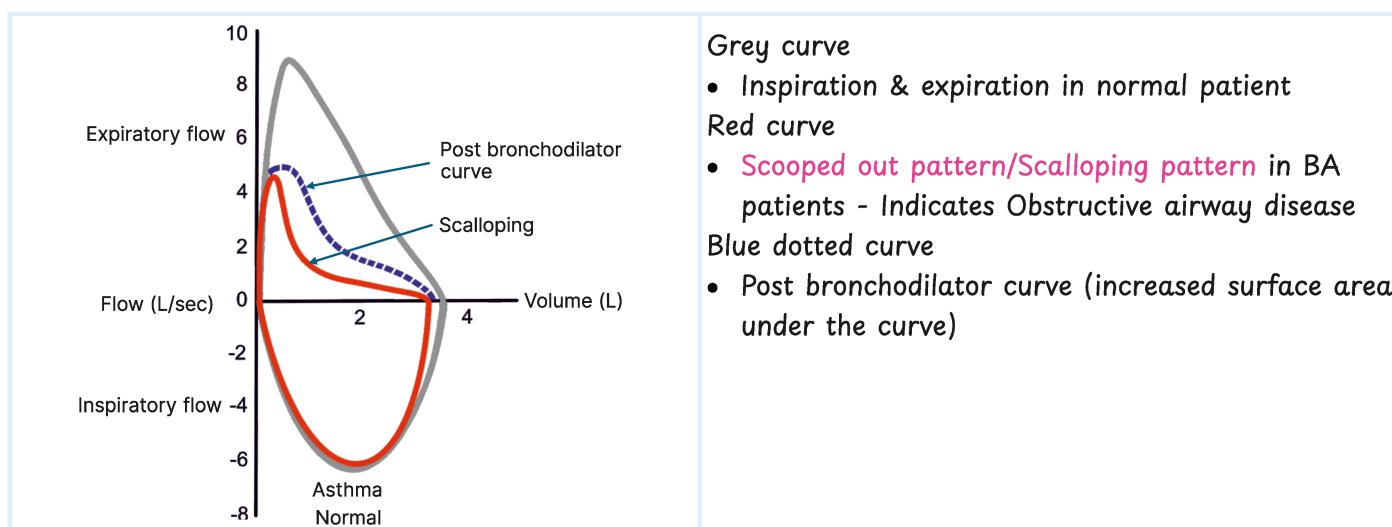
CLINICAL FEATURES SUGGESTIVE OF REACTIVE AIRWAY DISEASE/ASTHMA

- Recurrent wheezing with Shortness of Breath
- Cough on exposure to irritants/sudden change in temperature
- Symptoms worsen at night time/on awakening
- Assessment for comorbidity if attacks occur despite moderate inhaled corticosteroids + LABA
- Asses for drug intake - ACE inhibitors, β blockers, COX-1 inhibitors
- Asses for nasal polyps & allergic rhinitis

DIAGNOSIS

00:39:16

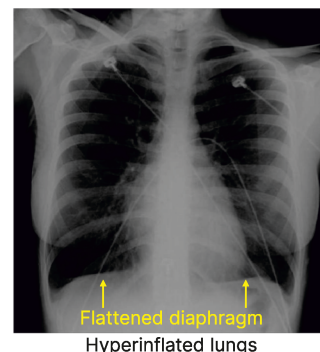
- IOC for BA- Spirometry



- Flow volume curve (Pre & post bronchodilator)
 - FEV1 increment of $>12\%$ after bronchodilator
 - Timed vital capacity: FEV1/FVC ratio <0.7
- Diurnal variation in PEFR $>20\%$
- Methacholine challenge test: FEV1 reduced by $>20\%$ (mannitol/3% saline nebulization)
- Fe NO >50 ppb in **adults** & >20 ppb in **children**
- Eosinophil count (value >300 cell/cumm) indicates response to anti IL-5 biologics
- IgE elevation >1000 IU/ml with brown sputum plugs suggestive of **Allergic bronchopulmonary aspergilloma**

Important Information

- Timed vital capacity >0.8 indicates Restrictive lung disease
- X-ray
 - Normal between attacks
 - Abnormality seen during attack- Hyperinflation (flattening of Diaphragm)
- Air trapping- $\uparrow$ RV, $\uparrow$ TLC



ASTHMA EXACERBATION SEVERITY

	MILD	SEVERE	IMMINENT RESPIRATORY ARREST
TALKS IN	Sentences	Words, monosyllables	Can't talk
ALERTNESS	Agitation $\pm$	Agitation	Drowsy
WHEEZING	Loud rhonchi in expiration	Loud rhonchi in both inspiration and expiration	Silent chest
PEF/FEV₁	70-80%	<40%	<25%
PaO₂	>60 mmHg	<60 mmHg, SpO ₂ <90%	<60 mmHg
PULSE/MINUTE	<100	>120	Bradycardia
PULSUS PARADOXUS	Absent	May be present	Absence suggests respiratory muscle fatigue, shallow breathing

FUNCTIONAL ASSESSMENT

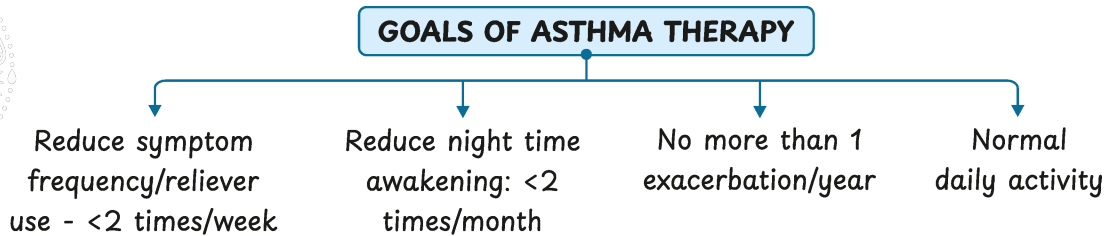
PEF OR FEV₁ % PREDICTED OR % PERSONAL BEST	$\geq 70\%$	<40%	<25%
PaO₂ (ON AIR, MMHG)	Normal	<60: possible cyanosis	<60: possible cyanosis
PCO₂ (MMHG)	<42	≥ 42	≥ 42
SaO₂ (ON AIR)	>95%	<90%	<90%

MANAGEMENT

00:56:48

- **Mild to Moderate acute asthma attack**
 - β_2 agonist formoterol + Inhaled corticosteroids
 - Formoterol: LABA with rapid onset
 - Use of SABA alone- Not recommended
- **Severe acute asthma: Admit to urgent care centre**
 - Nebulization with Salbutamol
 - Humidified Oxygen-driven nebulization
 - Air-driven nebulization causes ventilation-perfusion mismatch
 - I/V Hydrocortisone: Inhibits Phospholipase A2 causing inhibition of leukotriene production
 - Anticholinergic - Ipratropium nebulization
 - Magnesium sulphate I/V - Mg²⁺ is antagonist of intracellular Ca⁺ inhibiting smooth muscle contraction





STRATEGY IN ASTHMA MANAGEMENT

- Mitigation: ↓ Allergen exposure
- Allergen immunotherapy
- Vaccines: Pneumococcal, COVID-19, Influenza
- Drugs

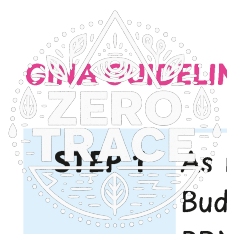
RELIEVER MEDICATIONS

- **Bronchodilators**
 - β_2 agonist
 - SABA: Salbutamol (4-6 hrs)
 - LABA: Formoterol (12 hrs)
 - Ultra LABA (used in COPD): Indacaterol (24 hrs)
 - S/E: Palpitations, Tremors, HTN, Hypokalemia, Type B Lactic acidosis
 - Anticholinergics: Ipratropium
 - Theophylline: Not used due to toxicity (Multifocal atrial tachycardia)

CONTROLLER MEDICATIONS

- **Inhaled corticosteroids / LABA**
 - **First line** therapy for persistent asthma
 - Decrease mortality
 - Rinse mouth with water after every use
 - Prophylaxis for *P. jirovecii* if using oral Corticosteroids >20 mg/day
- **Leukotriene modifiers -**
 - Montelukast (S/E - Suicidal ideation)
 - Zafirlukast, Cromolyn sodium
- **Biologics**
 - Anti IgE: Omalizumab
 - MOA: Prevent binding of IgE to Mast cells and Basophils
 - Used if IgE >30 IU/ml & symptomatic with high dose ICS + LABA daily
 - IL-5 active drugs
 - Mepolizumab
 - Reslizumab
 - Benralizumab- Blocks IL-5 receptors

} Anti IL-5 molecules
 - Anti IL-4, IL-13
 - Both of them act on common receptor: IL-4 α
 - Dupilumab
 - Used in Type 2 asthma phenotype patient who is symptomatic with high dose ICS + LABA & >2 exacerbations requiring oral Corticosteroids & Eosinophil count >300 cells/cu mm



GINA GUIDELINE FOR ASTHMA

STEP 1	As needed low dose ICS- Formoterol (e.g. Budesonide- Formoterol 160/4.5 µg 1 puff PRN)	Symptoms <2x/month No risk factors for exacerbation	AIR
STEP 2	As needed low dose ICS- Formoterol (used more frequently than step 1)	Symptoms >2x/month One or more exacerbation in past year	AIR
STEP 3	Daily low dose ICS- Formoterol + as needed ICS- Formoterol (same inhaler)	Symptoms most days Night waking ≥1x/week Frequent reliever use	MART
STEP 4	Daily medium dose ICS- Formoterol + as needed ICS- Formoterol (same inhaler)	Poor control on step 3 Frequent exacerbation	MART
STEP 5	High dose ICS- Formoterol (optional) + Add on therapies (e.g. Biologics)	Severe asthma Not controlled on step 4 despite good technique & adherence	MART + Add-on

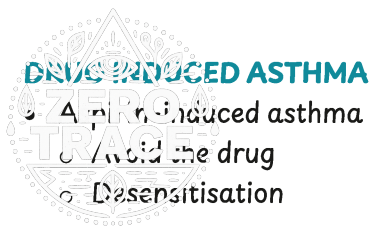
COMPARISON OF BIOLOGICS FOR ASTHMA

DRUG	TARGET	BEST FOR	TYPE 2 ASTHMA	ROUTE	DOSING FREQUENCY
OMALIZUMAB	IgE	Allergic asthma (high IgE)	Yes	SC	Every 2-4 weeks
MEPOLIZUMAB	IL-5	Eosinophilic asthma	Yes	SC	Every 4 weeks
RESLIZUMAB	IL-5	Eosinophilic asthma	Yes	IV	Every 4 weeks
BENRALIZUMAB	IL-5 receptor	Eosinophilic asthma (fast eosinophil depletion)	Yes	SC	Every 4-8 weeks
DUPIUMAB	IL-4 & IL-3	Allergic + Eosinophilic asthma	Yes	SC	Every 2 weeks
TEZEPELUMAB	TSLP (upstream inhibitor)	Both Type 2 & Non type 2 asthma	Yes	SC	Every 4 weeks

EXERCISE INDUCED ASTHMA

01:17:44

- If on occasional exercise: ICS/ LABA is used
- If on regular exercise: LTRA (Leukotriene receptor antagonists)



HYPERSENSITIVITY PNEUMONITIS

01:19:00

- Non asthmatic extrinsic allergic alveolitis due to inhalational exposure to antigens
- Specific circulating antibodies of IgG class
- Examples
 - Farmer lung (Barn hay- Thermophilic actinomycetes)
 - Bird fancier lung
 - Hot tub lung
 - Woodworker's lung
- Diagnosis of HP is based on clinical prediction rule (SOB, B/L crepts)
- T/t - Steroids



- Q. A 14-year-old boy is having asthma attacks since last night. He appears anxious, can talk in monosyllables and has RR 40/min with use of accessory muscles of respiration. Loud rhonchi are heard in both lung fields. SaO_2 is 85% at room air. Which of the following will not be useful for management of this patient?
- Terbutaline subcutaneous
 - Magnesium sulphate
 - Aminophylline intravenous
 - Salbutamol nebulization

Ans (c)

- Q. Analyse the given spirometry report and comment on diagnosis in this construction worker

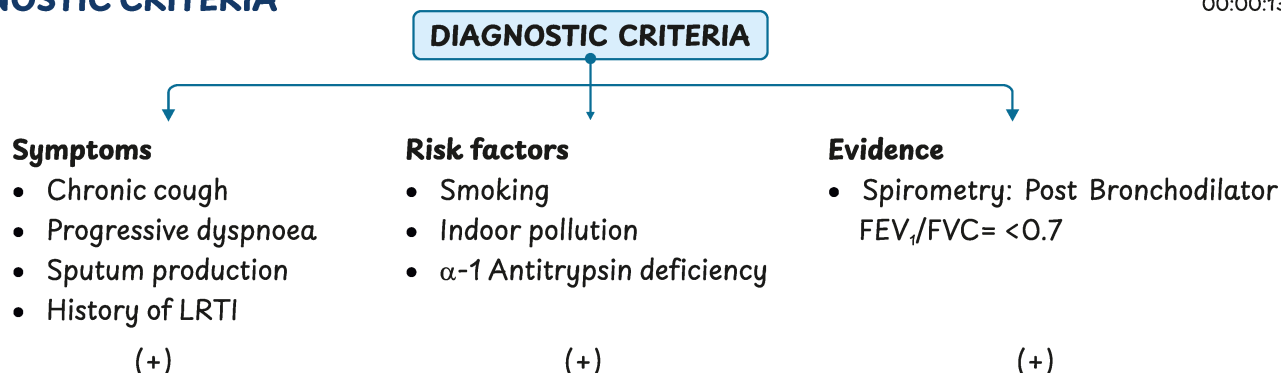
PARAMETER	PRE-BRONCHODILATOR	POST-BRONCHODILATOR
FEV ₁ (L)	2.00 L	2.30 L
FVC (L)	3.50 L	3.80 L
FEV ₁ /FVC Ratio (%)	57.1%	60.5%
PEF (Peak Expiratory Flow) (L/s)	5.0 L/s	5.5 L/s
FEF _{25-75%} (L/s)	1.80 L/s	2.10 L/s

- Asthma
- COPD
- I.P.F
- Silicosis

Ans (a)

DIAGNOSTIC CRITERIA

00:00:13



A-1 ANTITRYPSIN DEFICIENCY

- SERPINA 1 gene defect** (Chromosome 14)
- Autosomal codominant transmission
- Z allele: Marked reduction
- Null allele: Complete absence → Panacinar emphysema
- T/t: Weekly infusions of α -1 Antitrypsin

ETIOTYPES OF COPD

00:06:15

COPD-G	Genetic cause
COPD-C	Cigarette smoking
COPD-P	Pollution
COPD-I	Childhood infections
COPD-D	Developmental lung defects
COPD-A	Coexistent with asthma

GOLD GRADING AND SEVERITY OF AIRFLOW OBSTRUCTION

00:07:09

GRADE	SEVERITY OF OBSTRUCTION	FEV ₁ /FVC RATIO	FEV ₁
GOLD 1	Mild	<0.7	$>80\%$ of predicted value
GOLD 2	Moderate		50-79%
GOLD 3	Severe		30-49%
GOLD 4	Very severe		$<30\%$



DEGREE OF BREATHLESSNESS RELATED TO ACTIVITY	
0	Breathlessness only on strenuous exercise
1	Breathlessness on walking inclined / fast
2	Walks slowly as compared to peers, because of breathlessness
3	Stops for breath after walking about 100 yards or after a few minutes
4	Too breathless to leave the house/ change clothes (Grade 3 according to NYHA classification of breathlessness)

COPD ASSESSMENT TEST

00:14:40

I never cough	0 1 2 3 4 5	I cough all the time	
I have no phlegm (mucus) in my chest at all	0 1 2 3 4 5	My chest is completely full of phlegm (mucus)	
My chest does not feel tight at all	0 1 2 3 4 5	My chest feels very tight	
When I walk up a hill or one flight of stairs I am not breathless	0 1 2 3 4 5	When I walk up a hill or one flight of stairs I am very breathless	
I am not limited doing any activities at home	0 1 2 3 4 5	I am very limited doing activities at home	
I am confident leaving my home despite my lung condition	0 1 2 3 4 5	I am not all confident leaving my home because of my lung condition	

- Contains 8 questions
- Minimum score: 0
- Maximum score: 40 (5 per question)

GOLD ABE ASSESSMENT TOOL

- Used to create treatment groups

EXACERBATION HISTORY	TREATMENT GROUP	
> 2 moderate exacerbations > 1 hospitalization	E	
0-1 moderate exacerbation	A	B
	mMRC: 0-1 CAT:<10	mMRC: 2 CAT:>10